



Garren turbidity 2023-4

A report on behalf of the Wye Salmon Association.

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An analysis of nephelometric turbidity data from the Garren sub catchment between July-23 and July-24.

Introduction

The Garren Brook suffers from permanent turbidity. Even on its bones, this river does not run clear. The Environment Agency (EA) view is that this problem is due to fine clay particles originating from the sandstone bedrock, which are brought into suspension by agricultural land use practices. Frequently cited examples include run off from 'bare soil' crops such as potatoes and maize during wet weather. Livestock access to the streams could also be a factor, as could bankside erosion. Whatever the reasons, the persistent turbidity is implicated in the various ecological woes of the sub catchment: lack of macrophytes, lack of fish, etc.

Our earlier report on the Garren Brook (Green, 2022a) included an attempt to evaluate variation of turbidity around the sub catchment, based on subjective assessment of colour by our citizen scientist, Peter Hunt. Broadly, this analysis suggested that turbidity was generally higher in the lower sub catchment. But it was recognised that this is a very crude measure of turbidity. Peter has also used a Secchi tube. Beyond the obvious cumbersome nature of this method, he identified issues of precision resulting from the visual assessment. Target visibility was found to be dependent on ambient lighting conditions, for example. Something better was needed.

Available commercial equipment was felt to be prohibitively expensive for CS deployment. So WSA developed a 'home-brew', hand-held solution using the same nephelometric measurement principle as the sondes occasionally deployed by EA. A prototype unit was field tested by Peter during 2022-3. Following some design refinement, the first 'production' unit was given to Peter in July 2023. One year on, we have accumulated a good dataset to evaluate, this being the subject of this report.

Garren sub catchment

Overview

The Garren sub catchment was described in some detail in our earlier work (Green, 2022a), so here we shall recap just the more salient points. Figure 1 shows a topographical map of the sub catchment which covers 90 km² and is located a few km west of Ross on Wye in Herefordshire. The Garren joins the Wye just above Whitchurch. Roughly halfway along its course it is joined by the Gamber, whose confluence marks the separation of the three water bodies used for Water Framework Directive (WFD) purposes: the Gamber, upper Garren & lower Garren (delineated in red). Just downstream of the Gamber confluence, the Llanerch Brook joins from the west. The Luke Brook drains the extreme eastern side of the lower catchment and joins the Garren near Marstow.

The western side of the sub catchment is generally hillier than the eastern side, which drives a divergence in the predominant agriculture. The upper Garren has about 60% livestock and 30% arable landcover. The Gamber is the inverse, 60% arable and 30% livestock. The lower Garren is roughly evenly split.

This rural area contains only minor settlements; consequently, there are no major sewage treatment plants to consider. However, the Garren sub catchment does contain one of the highest densities of poultry to be found in Herefordshire, with 25 IPU's and 3 associated AD plants. As we showed in our Garren soils survey (Green, 2022b), there is a lot of poultry manure being dumped on the fields of this sub catchment.

Sampling locations

Peter Hunt has been monitoring 11 sites within the Garren sub catchment, plus a site on the Daff y Nant Brook – that's the stream to the south of the map, entering the Wye at Whitchurch. These are shown as yellow circles in figure 1. As always, locations are largely determined by road access but, broadly, there are two sites per water body, plus sites on key tributaries as outlined in Table 1.



Figure 1: the Garren sub catchment. Red outlines show the three WFD water bodies; yellow circles are WSA monitoring sites, with WyeViz reference numbers.

Reference	Site name	Description
<i>Upper Garren:</i>		
WSA104	Garren at Orcop	Water body upstream site
WSA103	Garren at Llangunnoch Bridge	Water body downstream site
WSA106	Brooklands Farm Stream	Small tributary stream draining 2 IPU and 1 AD plant
WSA152	Great Ynys Farm Stream	Small tributary stream draining 1 IPU and 1 AD plant
<i>Gamber:</i>		
WSA136	Gamber at Audits Bridge	Water body upstream site
WSA111	Gamber at Tretire	Water body downstream site
WSA101	Trevase Farm Stream	Tributary stream draining 1 IPU
<i>Lower Garren:</i>		
WSA102	Garren at Langstone Court	Water body upstream site, below Gamber confluence
WSA125	Garren at Packsaddle Bridge	Water body downstream site, just above Wye confluence
WSA139	Llanerch Brook at Panbrook Bridge	Tributary joining Garren below Langstone Court
WSA098	Luke Brook at Brelston	Tributary joining Garren above Packsaddle Bridge
<i>Daff y Nant:</i>		
WSA124	Daff y Nant Brook	Wye tributary flowing through Whitchurch

Table 1: summary of WSA sampling sites in and proximate to the Garren sub catchment.

Results and discussion

Basic statistics

The data set comprises 44-51 samples at each of the primary Garren and Gamber sites, spanning the period 17-Jul-23 to 15-Jul-24. The Trevase Farm stream was sampled 39 times, and the Great Ynys Farm stream was sampled 19 times commencing on 23-Oct-23.

To aid interpretation, the data is presented on a schematic diagram of the catchment. Figure 2 shows the annual mean turbidity at each site.

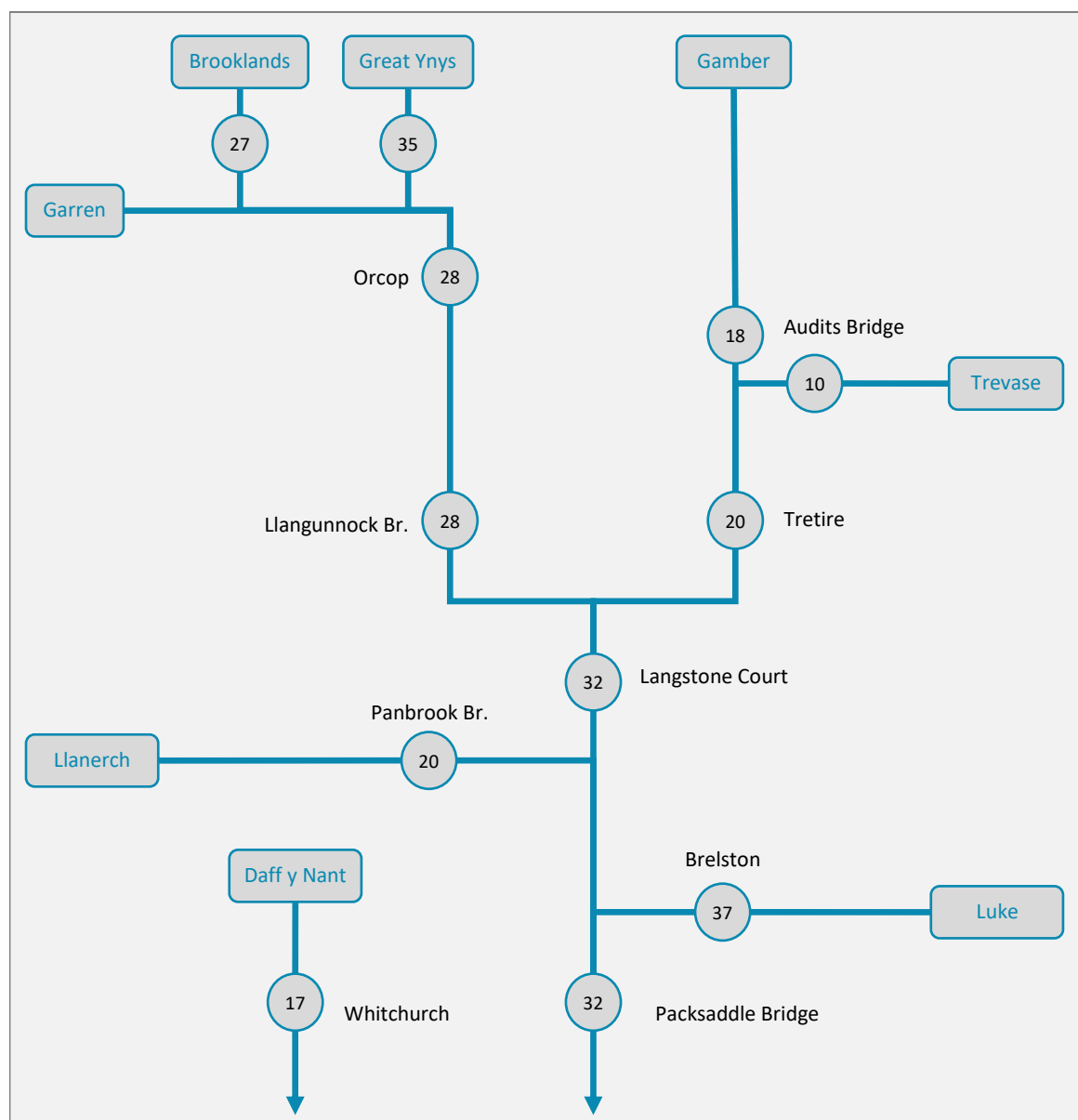


Figure 2: annual mean turbidity at monitored sites

It is immediately apparent that our earlier analysis of subjective colour assessment data is not borne out by quantitative measurement. The turbidity is not generally higher in the lower catchment; a reasonable interpretation being that the subjective data is influenced by the generally larger depths in the lower catchment reducing visibility of bottom features. The mean turbidity data presented in figure 2 suggest that the Garren and Luke Brook have higher mean turbidity than the Gamber, Llanerch and Daff y Nant brooks. Moreover, the mean turbidity in the Garren is broadly uniform throughout its length. The two monitored tributaries on the Garren above Orcop also have high turbidity. Trend charts for these sites can be found in WyeViz, so are not replicated here.

Flow dependency

The EA monitor flow on the Garren at Marstow, just upstream of Packsaddle Bridge (see figure 1). Mean daily flow data for this location has been downloaded from the [EA Hydrology Data Explorer](#) and matched up to the turbidity sampling dates. Figure 3 shows the relationship between turbidity and volumetric flowrate at Packsaddle Bridge.

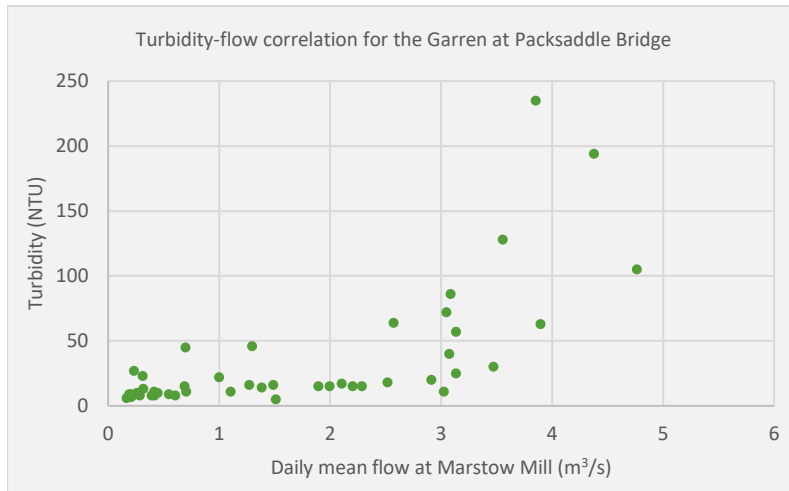


Figure 3: scatter graph showing turbidity at Packsaddle Bridge versus volumetric flowrate at Marstow Mill.

Annual mean flow (AMF) at Marstow Mill is $1.02 \text{ m}^3/\text{s}$, so we can approximately read the horizontal axis as multiples of AMF. The chart shows a reasonably well-behaved linear relationship up to $2 \text{ m}^3/\text{s}$, other than a few outliers, which we shall return to. Between $2 \text{ m}^3/\text{s}$ and $3 \text{ m}^3/\text{s}$, there is a strong inflexion with turbidity rising more rapidly at higher flowrates; there is also a significant increase in spread. All sites, except for Trevase Farm Stream, show a similar inflexion at the same point.

Low flow

It is instructive to examine the low flow region in more detail, as shown in figure 4.

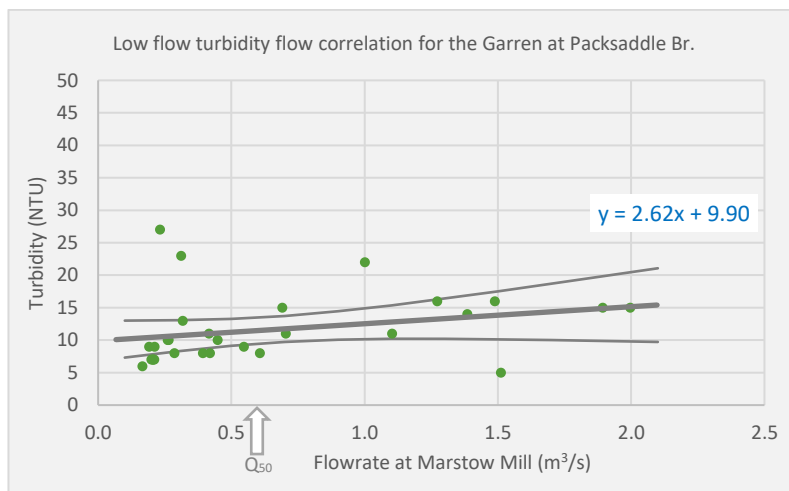


Figure 4: scatter graph showing turbidity at Packsaddle Bridge versus volumetric flowrate at Marstow Mill, for flowrates $\leq 2 \text{ m}^3/\text{s}$. The two outliers have also been removed. The regression line and 95% confidence limits are shown in grey. The white box contains the equation of the regression line. The white arrow indicates the Q_{50} flowrate of $0.61 \text{ m}^3/\text{s}$.

The regression line shows the approximate relationship to flowrate. This suggests two useful parameters: the gradient and the turbidity at Q_{50} flowrate¹ ($0.61 \text{ m}^3/\text{s}$) – the intercept at zero flowrate being a rather nonsensical concept.

Performing the same low flow regression process for each site yields the following results.

¹ Q_{50} is the flowrate which is exceeded for 50% of the time. It is therefore roughly what an observer might consider 'normal flow'.

Reference	Site name	Gradient	Turbidity at Q ₅₀ (NTU)	Min. turbidity (NTU)
<i>Upper Garren:</i>				
WSA104	Garren at Orcop	2.65	12.13	4
WSA103	Garren at Llangunnoch Bridge	1.32	10.62	3
WSA106	Brooklands Farm Stream	3.34	15.01	3
WSA152	Great Ynys Farm Stream	-2.05	11.46	5
<i>Gamber:</i>				
WSA136	Gamber at Audits Bridge	2.80	8.45	2
WSA111	Gamber at Tretire	1.41	10.22	5
WSA101	Trevase Farm Stream	-1.52	9.00	3
<i>Lower Garren:</i>				
WSA102	Garren at Langstone Court	1.73	14.55	5
WSA125	Garren at Packsaddle Bridge	2.62	11.49	5
WSA139	Llanerch Brook at Panbrook Bridge	-0.29	10.06	2
WSA098	Luke Brook at Brelston	5.50	11.74	4
<i>Daff y Nant:</i>				
WSA124	Daff y Nant Brook	0.82	9.08	3

Table 2: columns 3-4 list regression parameters for dates where the Marstow Mill flowrate is below 2 m³/s. Column 5 shows the minimum recorded turbidity at each location.

The flowrate dependencies (gradients) broadly lie in the range 1 to 3.5. The Luke Brook is an outlier at 5.5; a reason for this is not immediately obvious. Both the Great Ynys and Trevase Farm streams have strong negative gradients, which would seem to imply a fixed flow, anthropogenic source of turbidity. Not inconceivable, passing close to intensive livestock operations.

Average turbidity at Q₅₀ generally falls in the range 9 to 15 NTU. This is extremely high and is consistent with the known issues of this sub catchment. Also listed in table 2 are the minimum recorded turbidity values at each site. This shows that the upper Garren, Gamber & Llanerch Brook, for example, can run clear (2-3 NTU, highlighted green in Table 2); but they don't do it very often. Importantly, these low turbidity days do not coincide with the lowest flowrates and have no obvious seasonality. The lower Garren and Luke Brook do not seem to fall below 4-5 NTU.

The data spread around the regression line seen in figure 4 (and which is typical of all sites) shows that the variability in low flow turbidity is not primarily driven by flowrate. Other factors are clearly at play². For the Garren at Packsaddle bridge, the low flow turbidity spans 5 – 46 NTU, only 5 NTU of which is explained by flowrate (the vertical range of the regression line in figure 4). The clear implication of this observation is that the persistent turbidity in the Garren catchment is not the result of flowrate driven processes: ongoing bankside erosion or pick-up of mineral particles from the bed.

Figure 5 overleaf, presents the average turbidity at Q₅₀ on the catchment schematic. The upper Garren again has slightly higher turbidity than the Gamber. The tributary streams at Great Ynys Farm and Brooklands Farm also have high turbidity. Note the significantly lower number on the Trevase Farm stream. Below the Gamber confluence at Langstone Court, we have an anomalously high value, suggesting that a specific local problem exists below Llangunnoch and Tretire. This increase has dissipated by the time the Garren reached Packsaddle Bridge.

The higher low flow turbidity on the upper Garren compared to the Gamber correlates with two associated factors. Firstly, the upper Garren is hillier than the Gamber which may result in more energetic drainage pathways, although it is difficult to see this as a major factor in dry weather. Secondly, the upper Garren is dominated by livestock farming, as opposed to arable farming.

² The R² values for the regression lines are typically below 0.1, which indicates that most of the data variance is not being explained by flowrate.

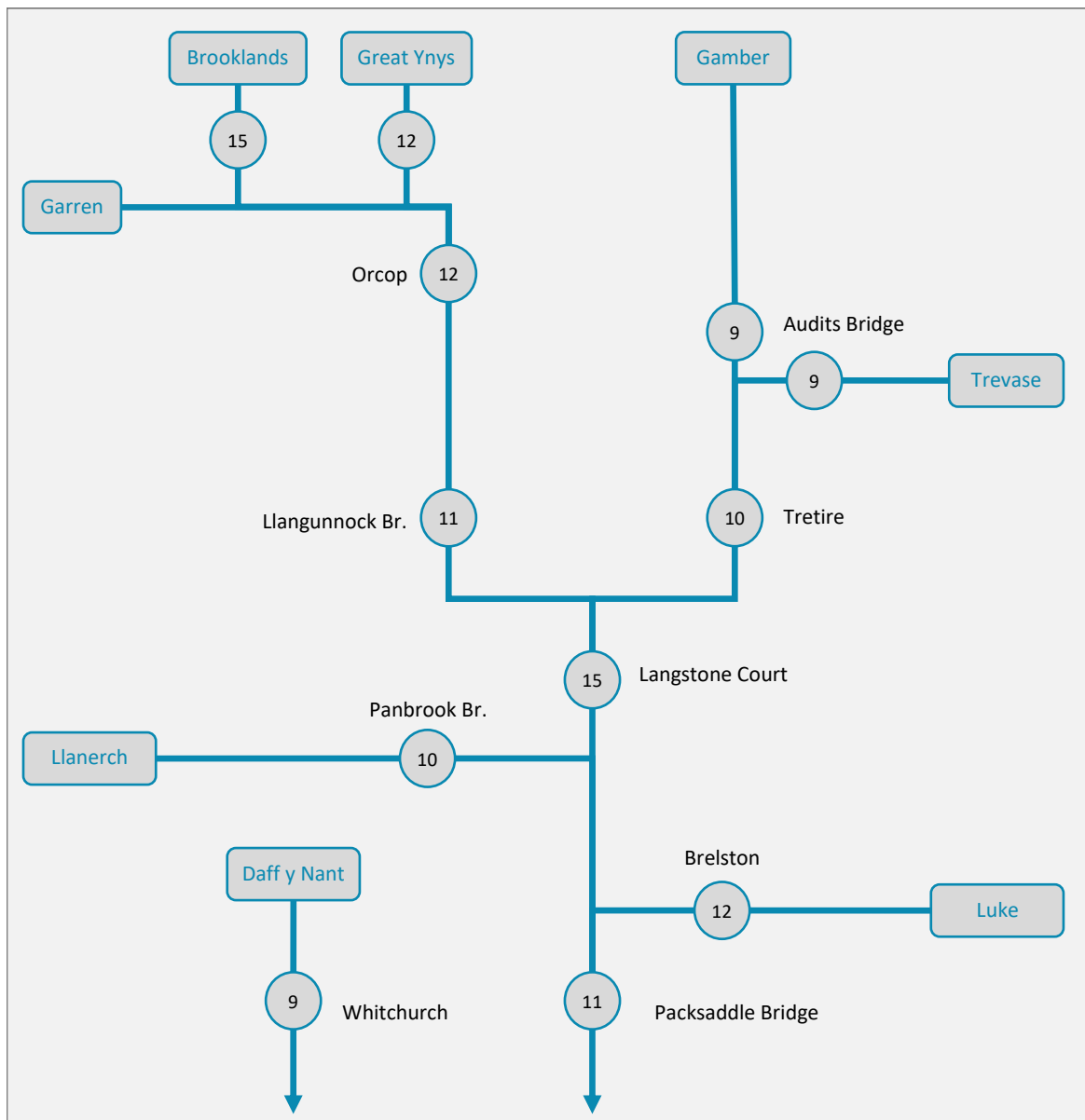


Figure 5: low flow turbidity at monitored sites, using regression derived mean turbidity at Q_{50} .

High flow

The low flow regime examined above for the most part approximates a steady-state situation. In contrast, the high flow regime is essentially transient in nature. Referring to figure 3, the turbidity data at flowrates above the inflection point at around $2.5 \text{ m}^3/\text{s}$ shows (a) a rapid increase in gradient, and (b) a significant increase in spread. The former arises because, during heavy rainfall, additional pathways turn on, often across the surface, which can carry larger quantities of soil particles than the groundwater pathways prevalent in dry weather. The increase in spread arises in part because these surface pathways are the first to turn off when the rain stops, and the river begins to fall. Thus, the turbidity on a rising river is generally higher than the turbidity on a falling river at the same flowrate. This phenomenon is referred to as hysteresis. Figure 6 overleaf shows the high flow region for the Garren at Packsaddle bridge. It shows a distinctly bimodal spread in turbidity data, which can be explained by such a hysteresis effect. This pattern is present at all sites on the upper Garren, lower Garren, Gamber, Llanerch and Luke brooks. Only Trevase Farm and Brooklands Farm streams do not show significant hysteresis.

The regression line on figure 6, although statistically dubious for bimodal data, does give us a means of parameterising each site's high flow data. Table 3 below shows the gradients of the regression lines at

each site, along with the derived value of turbidity at Q_{05} flowrate³ ($3.7 \text{ m}^3/\text{s}$) at Marstow Mill. The maximum recorded value is listed as a rough indication of spread, and thereby hysteresis.

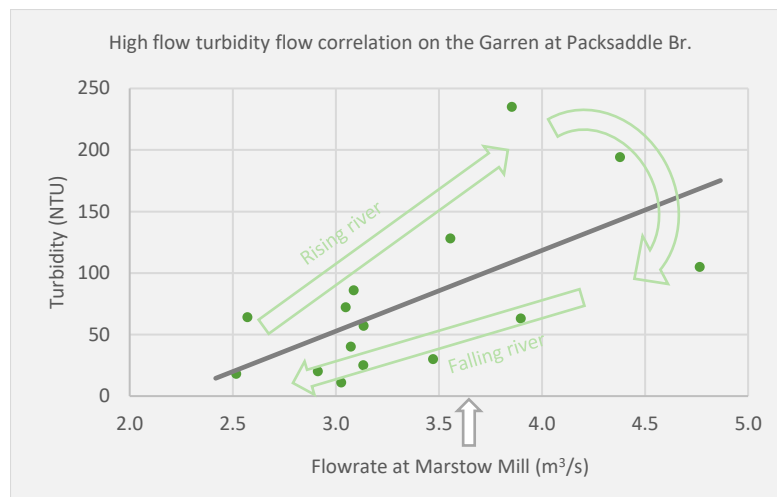


Figure 6: scatter graph showing turbidity at Packsaddle Bridge versus volumetric flowrate at Marstow Mill, for flowrates above $2.5 \text{ m}^3/\text{s}$. There is a clear suggestion of hysteresis on this data. The white arrow indicates the Q_{05} flowrate of $3.7 \text{ m}^3/\text{s}$

Reference	Site name	Gradient	Turbidity at Q_{05} (NTU)	Max. turbidity (NTU)
<i>Upper Garren:</i>				
WSA104	Garren at Orcop	15.35	58.14	131
WSA103	Garren at Llangunnock Bridge	37.72	80.27	240
WSA106	Brooklands Farm Stream	4.34	44.24	91
WSA152	Great Ynys Farm Stream	18.11	60.01	151
<i>Gamber:</i>				
WSA136	Gamber at Audits Bridge	14.39	35.64	96
WSA111	Gamber at Tretire	14.93	44.02	140
WSA101	Trevase Farm Stream	3.77	12.93	19
<i>Lower Garren:</i>				
WSA102	Garren at Langstone Court	64.72	94.97	240
WSA125	Garren at Packsaddle Bridge	65.62	100.65	235
WSA139	Llanerch Brook at Panbrook Bridge	15.63	49.28	205
WSA098	Luke Brook at Brelston	42.17	87.43	240
<i>Daff y Nant:</i>				
WSA124	Daff y Nant Brook	18.30	40.48	128

Table 3: columns 1-4 list regression parameters for dates where the Marstow Mill flowrate is above $2.5 \text{ m}^3/\text{s}$. Column 5 shows the maximum recorded turbidity at each location.

Looking at the gradients, we find that these are generally an order of magnitude higher than the low flow values, being indicative of the additional pathways referred to above. Exceptions are the Trevase Farm and Brooklands streams, neither of which have any inflexion in their graphs.

Average turbidity at Q_{05} is interesting. The Upper Garren is 58 NTU at Orcop, increasing to 80 NTU at Llangunnock Bridge. This trend continues in the lower Garren, with 65 NTU at Langstone Court and 101 NTU at Packsaddle Bridge. Luke Brook is also high. The Gamber runs cleaner than the upper Garren, with 36 NTU at Audits Bridge and 44 NTU at Tretire. This is most likely due to the flatter topography in the Gamber, and with slightly lower rainfall, providing fewer fast, surface pathways. But it does tend to question the often-heard assertion that bare arable fields are responsible for much of the soil load in rivers. We should perhaps also be looking at heavily compacted pasture on sloping topography as a source of wet-weather turbidity (Collins et al., 2021).

³ Q_{05} is analogous to Q_{50} , representing the flowrate which is exceeded only 5% of the time. It thus corresponds to 'high flow' conditions.

Maximum recorded turbidity requires a health warning. It is a single data point, as opposed to a statistic based on all data. As such, we may have missed the peak, particularly on the tributary streams which will rise and fall faster than the main stems of the Garren and Gamber. That said, the final column tells a similar story, with the upper Garren showing 131 NTU at Orcop and 240 NTU at Llangunnock. Both lower Garren sites have also seen around 240 NTU, as has the Luke Brook. Highest recorded turbidity on the Gamber is 96 NTU at Audits Bridge and 140 NTU at Tretire. So, this metric also shows the Gamber to be cleaner than the Garren.

Outliers

Odd outliers in the low flow regime (see figure 3) require explanation. These are typical of the data for most sites and could be due to natural causes such as recent rainfall, or they could represent anthropogenic point pollution. Picking these out manually from the dataset and collating their dates yields the data shown in table 4 below.

		Date	24/7/23	16/10/23	23/10/23	6/11/23	20/11/23	27/5/24
		Flowrate (m3/s)	0.318	0.233	0.699	1.298	1.997	1.001
		Trend	Steady	Falling	Steady	Falling	Steady	Falling
		Rain recorded?	Yes	No	No	No	Yes	Yes
Reference	Site name							
Upper Garren:								
WSA104	Garren at Orcop				23		143	34
WSA103	Garren at Llangunnock Bridge			31	21	19		
WSA106	Brooklands Farm Stream	52						37
WSA152	Great Ynys Farm Stream				49		91	
Gamber:								
WSA136	Gamber at Audits Bridge		19			17	23	
WSA111	Gamber at Tretire							22
WSA101	Trevase Farm Stream							
Lower Garren:								
WSA102	Garren at Langstone Court		32		30	32		
WSA125	Garren at Packsaddle Bridge		27		45	46		22
WSA139	Llanerch Brook at Panbrook Br.	30				21		
WSA098	Luke Brook at Brelston				46	240		
Daff y Nant:								
WSA124	Daff y Nant Brook	27			20	21		

Table 4: Occurrence of outliers in the low flow dataset. Numbers are recorded turbidity (NTU). Almost all outliers fall on one of six dates. Also recorded are the river flow and recorded weather conditions.

Two outliers were unique: one at Packsaddle Bridge on 17/7/23 and one on the Brooklands Farm stream on 17/6/24. All the others fell on one of six dates, as shown in the table. On each date, turbidity spikes were recorded at multiple locations having no surface water connection. So, for example, spikes were observed on 20/11/23 on the Garren at Orcop and nearby on the Great Ynys stream. A simultaneous spike also occurred on the Gamber at Audits Bridge, physically not far away, but hydraulically unconnected. And, indeed, we see that rain was recorded in the previous 24 hours. No rain was recorded on the 23/10/23 data, nevertheless, the wide geographical spread is strong evidence of a natural cause. It also seems unlikely that transient algal blooms would arise simultaneously on the Gamber and Garren. In summary, none of the tabulated outliers appear to have an anthropogenic origin.

Conclusion

- The WSA turbidity meter has proved to be a useful tool for understanding the rivers. It has shown the intrinsic weakness of visual assessment, and offers superior precision, lower limit of detection and improved convenience compared to a Secchi tube.
- Application on the Garren sub catchment has benefitted greatly from the extent of the dataset, both in terms of its spatial spread and in terms of the reliable weekly sampling. (Well done, Peter!)
- Analysing turbidity data as a function of river discharge is a more powerful analytical approach than, say, simple time trend plots. It is useful to consider low flow and high flow regimes separately. Linear regression is then a useful tool to derive relevant summary metrics: the turbidity at Q_{50} characterising low flows and that at Q_{05} characterising high flows. Whilst the use of daily flow data is strictly only valid for sites close to the measuring station, it has been found to be a reasonable approximation for all sites in this relatively small sub catchment.
- Conclusions with respect to the Garren sub catchment are as follow:
 - Low flow turbidity values, characterised by the regression value at Q_{50} show values in the range 9 – 15 NTU throughout the upper Garren, lower Garren & Gamber systems and on the adjacent Daff y Nant. These are very high values by the standards of the wider Wye catchment.
 - Regression analysis shows that only a small fraction of the low flow data variance is explained by flowrate. This suggests that natural kinetic effects, such as pick up of particles from the bed, or bank erosion, are not the primary driver of low flow turbidity. Other factors must be at play.
 - If we accept the EA belief that the turbidity is mineral in origin, then this mineral load is most likely being mechanically agitated in-stream by livestock and/or is entering from land drains, potentially having anthropogenic drivers (misconnections, etc.). The observation that both the upper Garren and Gamber can run at 2-3 NTU on odd occasions supports the hypothesis that we are dealing with episodic sources such as these.
 - The observation that low flow turbidity is lower on the Gamber than on the upper Garren and lower Garren would tend to implicate livestock farming activities.
 - The anomalously high turbidity recorded at Langstone Court would suggest a specific, local source somewhere downstream of Llangunnock on the Garren and of Tretire on the Gamber.
 - The negative regression gradients found on the Great Ynys Farm and Trevase Farm streams are suggestive of fixed flow pollution sources.
 - High flow turbidity values, characterised by the regression value at Q_{05} , also show higher values on the upper Garren and lower Garren than on the Gamber. This is likely to be due to the hillier terrain in combination with livestock farming practices. Arable farming evidently puts less soil into the Gamber during winter floods than livestock farming puts into the upper Garren.
 - Maximum turbidity values tell the same story, with higher values recorded on the Upper Garren, Lower Garren and Luke Brook than on the Gamber.

Recommended actions

- WSA: consider expansion of this approach to other sub catchments. The Dore is an obvious contender since the team there already have several turbidity meters. We might also trial the application of turbidity meters to hot spot investigations, which have so far focussed on nutrients.
- Wye Alliance: consider the application of nephelometric turbidity measurement within the wider CS community and perhaps its further integration into the WyeViz platform.
- Wye Catchment Partnership: consider how the farming community can be engaged in finding solutions to elevated turbidity, which is essentially an unintended environmental consequence of modern agricultural practices.
- EA: consider how to use this new CS data to help improve river water quality, for example to help target farm advisory visits, etc.

References

Collins, A. L., Zhang, Y., Upadhayay, H. R., Pulley, S., Granger, S. J., Harris, P., Sint, H., & Griffith, B. (2021). Current advisory interventions for grazing ruminant farming cannot close exceedance of modern background sediment loss – Assessment using an instrumented farm platform and modelled scaling out. *Environmental Science and Policy*, 116, 114–127. <https://doi.org/10.1016/j.envsci.2020.11.004>

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